USA Master’s — Your Personalised Booklet

How to get into the USA for a master’s — written for one applicant, not for everyone.

For: **Muhammad Ismail** — BS Energy Engineering, UET Peshawar (USPCAS-E), Oct 2023 – Oct 2027, GPA 3.46, from Charsadda, Khyber Pakhtunkhwa, Pakistan
Target: **MS in the USA, starting Fall 2028** (September 2028)
November 2026. Companion to the Erasmus Mundus v2 booklet — same method, different country, different rules.

The one thing to understand before anything else. The USA has no Erasmus-style scholarship. There are two separate routes, and this booklet covers both, because you should run them in parallel: **(1) Fulbright via USEFP** — fully funded, very competitive, and you cannot choose the university; **(2) direct applications to US universities** — you choose the university, but for a master’s the funding usually comes from a professor’s research or teaching assistantship (RA/TA), not from the university handing you a scholarship. That is why professor outreach matters far more for the USA than for Erasmus — and why this booklet has a whole chapter on it. The two routes have different deadlines, different essays, and different tests. Do not mix them up.

The one rule that runs through the whole booklet. Never invent a fact to make an application look stronger. No invented FYP site data, no invented job, no inflated software skill, no claimed test score you do not have, no claimed publication. US universities check transcripts and references, and one invented claim can end an otherwise strong application. Where your record is still in progress — Python, HOMER Pro, MATLAB, GIS — say so, and show the work you are doing to close the gap.

Timing warning — the USA is earlier than Erasmus. Your Erasmus applications open in October 2027. Your Fulbright application is due around **1 April 2027** — six months earlier, before you graduate, and the GRE must be taken before that deadline. If you follow the Erasmus timeline for the USA, you will miss Fulbright entirely. Chapter 13 gives you the USA timeline; read it before you plan anything else.

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1. Your profile, honestly assessed

The same verified profile as the Erasmus booklet — but read through an American lens. US admissions committees weight research fit, test scores, and references heavily, and they fund master's students through professors, not through a central scholarship.

Item	The fact	What it means for the USA
Degree	BS Energy Engineering, UET Peshawar (USPCAS-E), Oct 2023 – Oct 2027	16 years of education — meets Fulbright's requirement. Final-year students may apply to Fulbright if the final transcript can be provided by end-December (pattern from the 2027 cycle — confirm for 2028 when the call opens).
GPA	3.46 (on a 4.0 scale)	Clears every stated 3.0 minimum in Chapter 5. Above Georgia Tech's 3.3 "desirable minimum" for its cybersecurity

		MS. Just below Michigan ECE's typical 3.5 bar for its MS ECE — avoid that programme, use the MEng instead. Your projects must do the extra work.
FYP	Low-head hydropower from KP irrigation network — 12 canal sites, supervisor Prof. Dr. Farid Ullah Khan, group: M. Umar Shah, M. Ismail, Abdul Majid	Your strongest asset for both routes. Thesis-based MS programmes and professors funding hydropower/energy-systems research want exactly this.
Internships	WAPDA Tarbela (4 weeks: electrical maintenance, switchyard, turbines, generators); Tarbela 5th Extension, PCCCL (project intern since Sept 2026, 3 months, hybrid)	Field + plant experience. Note: Berkeley ERG prefers ≥2 years' work experience — address it head-on in that SOP, do not hide it.
Floating PV	UET research contributor; 120 weekdays of field data; comparative workbook; manuscript in preparation	Research experience — the single most valuable line for RA funding. "In preparation" is the honest status; never call it published or submitted.
Energiscope	Public web platform, Vercel; GitHub: engr-energy/pakistan-energy-dashboard	See the red box below — the CV data claim is wrong and must be fixed before any US application.
HOMER Pro	Gulabad, Charsadda case: PV–hydro–battery hybrid, NASA POWER data; load + solar assessment done	Techno-economic modelling proof — maps to energy-systems and energy-economics programmes.
Python / MATLAB / GIS	Python developing (Lesson 01 done); MATLAB R2025b, Simscape licence check pending; QGIS with Copernicus GLO-30 DEM, corridor clip in progress	Say "developing / in progress", never "proficient". A professor funding an RA wants to see the work, not the adjective.
Tests	No GRE score. No TOEFL/IELTS score.	GRE is **mandatory for Fulbright** (minimum 145 per section) and required at some direct programmes. Plan in Chapter 11: GRE by March 2027, TOEFL by mid-2027.
Publications	None	Normal for a bachelor's applicant. No opened source said "no publications = no RA". Do not compensate by inflating the floating-PV manuscript.
Leadership	GIZ Youth Ambassador (2023–2025); AEE UET member/	

	organiser (2023–present); SESA Campus Lead (2026–present)	Useful for Fulbright (leadership and service matter there) and for fellowship nominations. Keep it short in the résumé.
Other tracks	Mitacs GRI first-phase selection (professor matching); Erasmus Mundus booklet v2 (Sept 2028 target)	Keep all three tracks separate. Never use one as leverage for another, and never tell a US professor you are “also applying to Erasmus”.

> **FIX BEFORE YOU APPLY — Energiscope data claim.** Your refined CV says Energiscope is “built and deployed on real NEPRA generation data” with “10+ years of NEPRA generation data covering ~40 GW”. That is not true of the repository version: the repo contains only 6 years of sample data (2020–2025) hardcoded in app.js. US professors and Fulbright reviewers open GitHub links. **Steps:** (1) upload the 6 P0 replacement files from Energiscope_P0_Fixes.zip via GitHub web upload (November 2026, under an hour); (2) change the CV/résumé line to “public web platform visualising generation data (sample dataset 2020–2025; methods page in progress)”; (3) only after sourced NEPRA data is actually in the repo may you write “real NEPRA data” again.

What makes you competitive in the USA, specifically: (1) fieldwork on real water infrastructure — rare for an MS applicant; (2) a thesis-ready FYP with a real constraint (irrigation first) — exactly what a professor funds an RA to work on; (3) built, shipped things (Energiscope, digital twin, CanalPower PK) with links a professor can open; (4) a coherent story — decentralised renewable energy for places the grid serves poorly — that fits Fulbright’s “return and serve Pakistan” requirement perfectly.

2. Your story in one page

Same spine as the Erasmus booklet — one story, used everywhere, with the ending adjusted per route. For Fulbright, the ending must be explicit about returning to Pakistan; for direct applications, the ending must name the professor’s research area.

> **Where you come from.** You are from Charsadda, Khyber Pakhtunkhwa, where irrigation canals run past communities that the grid serves poorly. You grew up seeing water that does useful work for farmers — and nothing else. That is not a decorative opening; it is the reason your FYP exists.
>
> **What you did about it.** At UET Peshawar you chose Energy Engineering, and for your final-year project you took the hardest honest version of the question: can the unused head at canal falls become electricity — without taking water or timing away from irrigation? You walked the canals at Gulabad–Sardheri and Shabqadar, sat with irrigation officials at Warsak, and built a physics-first ranking ($P = \rho g Q H \eta$) with GIS screening, seasonal flows, and an irrigation veto. Alongside it you worked inside real hydropower — four weeks at WAPDA Tarbela, then the

Tarbela 5th Extension project — and you benchmarked floating solar against ground-mounted PV in the field for 120 weekdays.

What you build. Energiscope, a public energy-data platform. A working digital twin of your FYP canal network. CanalPower PK, an open screening tool for KP canals. Each one is small, open, and checkable — and each taught you to say "not measured yet" instead of inventing a number.

What you need next. A master's that teaches you to design decentralised energy systems properly — resource assessment, system modelling, grid integration, and the economics that decides whether a good engineering idea ever gets built.

What you will do after — Fulbright version. Return to Pakistan within 30 days of finishing, and serve Pakistan for at least two years — in hydropower and renewables project work or research — applying the same method to real sites. Fulbright requires this commitment; your story already lives it. **Direct-application version:** return to Pakistan's energy sector to work on decentralised renewables and grid integration, with the research skills from a thesis-based MS.

3. The two routes: Fulbright vs direct admission

	Route 1 — Fulbright (USEFP)	Route 2 — Direct application to universities
Who pays	Full scholarship: tuition, textbooks, airfare, living stipend, health insurance; plus visa fees, a computer and settling-in allowance	You, unless you win an RA/TA (tuition waiver + stipend) or a fellowship. Most MS students at most departments are expected to fund themselves at admission.
Who chooses the university	**Not you.** IIE placement professionals in New York match you to universities. You may note preferences; they are not guaranteed.	You choose — up to 6–8 universities from Chapter 5.
Deadline	~**1 April 2027** (pattern; 2028 call not yet announced — check from December 2026)	Mostly December 2027 – February 2028 (see Chapter 5 table)
Tests needed	GRE General **required**, minimum 145 per section, before the deadline. TOEFL only if shortlisted — IELTS is not mentioned on USEFP pages, do not assume it counts.	GRE: not required at 9 of the 10 shortlisted programmes (check each page). TOEFL iBT 85–100 or IELTS 6.5–7.0 typical; MIT TPP needs IELTS 7.5 and does not accept TOEFL.
Essays	Two essays: Study Objectives + Personal Statement (about 800 words	One SOP per university (500–1,000 words, programme-specific)

	each per USEFP's own instructions — quote those, not guidebooks)	
References	Three reference letters	Usually three LORs per university (same referees, adapted)
After the degree	**Must return to Pakistan within 30 days and serve ≥2 years** (J-1 212(e) rule). Cannot take H-1B, L-1, K visa or a green card during that time without a waiver — possible but difficult, never assume it.	F-1 visa; OPT work authorisation after the degree is possible under normal F-1 rules (check current rules at the time).
Application cost	Free to apply	$75–$114 per university (6 universities ≈ $450–$630)
Odds	~100 scholarships hoped for per cycle (2027 figure); thousands apply	Varies by programme; RA funding is the bottleneck, not admission alone

> **Run both routes in parallel — with one question to settle first.** Whether direct applications are permitted alongside a Fulbright application is **not addressed on USEFP's official pages**. Before you pay any direct-application fee, email USEFP (info@usefp.org) and ask exactly that question, in writing, and keep the reply. Do not assume. Everything else in this booklet assumes you run both — Fulbright first (April 2027), direct applications second (Oct 2027 – Jan 2028) — because the Fulbright deadline comes first and its GRE requirement covers both routes anyway.

4. Fulbright (USEFP) in full

Source: USEFP's official Fulbright Degree pages (opened and read, 5 November 2026): usefp.org/scholarships/fulbright-degree.cfm and /fulbright-degree-FAQ.cfm. Where this chapter says "pattern", the 2028 call is not yet published — re-check from December 2026.

4.1 Eligibility — checked against your profile

- **Education:** a four-year Bachelor's, or Bachelor's + Master's totalling 16 years of formal education from an accredited university. Your BS (Oct 2023 – Oct 2027) qualifies. Final-year students: the 2027-cycle FAQ says a student completing in 2026 could apply if the final transcript could be provided by end-December 2026. By the same pattern, you (completing Oct 2027) would need your final transcript by end-December 2027 for the 2028 cycle — **confirm the exact wording when the 2028 call opens**.
- **Nationality/residency:** Pakistani citizens with strong academic histories, residing in Pakistan throughout the process, committed to returning and serving Pakistan.

- **Work experience: not required** — unless you apply for Public Policy/ Administration or an MBA (MBA needs ≥2 years). For an Energy Engineering master's, your internships-only profile is not a barrier.
- **Fields:** all disciplines except clinical medicine. Applicants in "Sciences, Technology and AI, Trade and Economics, or Natural Resource Management are particularly encouraged" — Energy Engineering sits in the encouraged area.
- **Ineligible:** dual US/Pakistan nationals; anyone whose spouse or child over 18 is a US citizen/permanent resident; Fulbright/State/USAID employees and families; anyone who already held a Fulbright; anyone already studying in the US or on a work/residence/student visa in any country.

4.2 What it covers

Tuition, required textbooks, airfare, a living stipend, and health insurance — plus visa fees, a computer and settling-in allowance (USEFP FAQ). Awards "enable grantees to live at the same standard as most American graduate students." No dollar amounts are published for the stipend or allowances — **do not cite any figure as official**. Dependents are not funded. The application itself is free.

4.3 The 2-year return rule — understand it before you apply

Every selected candidate signs a contract binding them to return to Pakistan immediately after the programme (within 30 days) and serve for a period equivalent to the scholarship, minimum two years. This is the J-1 visa's 212(e) two-year home-residency requirement: during those two years you are not eligible for immigrant visas, permanent residence, or H, L, or K visas, and cannot change to most other US statuses, unless the requirement is waived. A waiver is legally possible (for example a No Objection Statement) but US-government financing counts strongly against granting one — possible, difficult, never assume it. Your story (Chapter 2) already commits to returning; write that commitment into both essays honestly, because Fulbright selects people who mean it.

4.4 The GRE — mandatory, and the one cost you can partly recover

- GRE General is **required for all applicants**. Minimum **145 on each section** under the current scoring system, "with a higher score on the section most applicable to your field" — for engineering, Quant carries more weight. Target: Quant 165+, Verbal 155+, Analytical Writing 4.0 (Chapter 11).
- Take it **before the application deadline**. Upload the score with the application; unofficial scores plus proof of registration are accepted if the official report is late.
- Send scores via ETS to **USEFP code 9388** and **IIE code 2326**.
- **Fee reimbursement:** if you study or teach at a public university and score **300+ combined (V+Q) in engineering/science**, attach a reimbursement request with the application. Up to 30 requests are honoured, first-come-first-served. UET Peshawar is public and Energy Engineering qualifies — so a 300+ score makes you eligible to request it. You still pay the fee yourself first (GRE fee $249, global rate).

4.5 How selection works — the dated steps

1. **~February 2027 (pattern):** the 2028 call opens. Check usefp.org/scholarships/fulbright-degree.cfm from December 2026. The 2027 cycle opened ~20 February 2026 with a 1 April 2026 deadline.
2. **~1 April 2027 (pattern):** application deadline. Package: online form, three reference letters, GRE score report, scanned transcripts including a grading-scale explanation (HEC-attested preferable). Incomplete applications are not considered; no paper applications.
3. **Round 1:** screening for completeness and minimum qualifications. **Round 2:** evaluation and shortlisting. **Round 3:** interviews, June–August (panels of US and Pakistani citizens). **Round 4:** Principal and Alternate nominees announced in September (email + official letter). Final approval by the Fulbright Foreign Scholarship Board in Washington DC.
4. **After selection:** university placement (by IIE — you cannot choose) continues until April; visa interviews June–August; departure July onwards.
5. **TOEFL:** only selected/shortlisted candidates take it. Do not pay for TOEFL before shortlisting — but be ready (Chapter 11).

Warning: a widely shared "Fulbright deadline 6 October 2026" is **not** the Pakistan deadline — it belongs to the Fulbright US Student Program for American citizens going abroad. Ignore it.

5. Your shortlist: 10 US universities

Target: 6–8 applications. Three Reach, four Target, three Likely. Every programme below was checked on its official page on 5 November 2026. All deadlines quoted are the Fall 2027 cycle as published — **Fall 2028 dates were not published anywhere**; expect a similar window and re-verify September–December 2027. Your GPA 3.46 clears every stated 3.0 minimum below.

The funding truth, before the list: only two programmes on this list have genuinely encouraging official funding statements for MS students — MIT TPP (research assistantships are the presumptive model) and Montana State (the majority of graduate students funded, MS auto-considered). Everywhere else, budget as self-funded and treat an RA/TA as a bonus. That is not pessimism; it is what the universities themselves publish.

5.1 Reach — 3 universities

R1. Stanford University — MS in Electrical Engineering

Programme: MS in Electrical Engineering, Department of Electrical Engineering. Coursework-based terminal MS (45 units, 1.5–2 years, autumn start only, no thesis). Fit via the EE research area "Energy, environment, and sustainability" / power electronics / smart grid — no dedicated hydropower track. **Fit for you:** FYP (variable, irrigation-constrained canal hydro) → generation, power electronics, grid integration; Tarbela switchyard/turbine/generator work → power-systems coursework; floating PV → renewables integration and solar-plus-storage. **Funding — honest flag:** largely self-

funded. The department states MS students are not eligible for departmental grants, scholarships or fellowships; RA/CA posts only after enrolment, after at least one quarter, in good standing. The full-funding route is **Knight-Hennessy Scholars** — a separate, fully funded application with its own deadline. **GRE:** not required. **Deadline:** December (1 Dec for Autumn 2027). **Page:** ee.stanford.edu/admissions/ms

R2. University of California, Berkeley — MS in Energy and Resources (ERG)

Programme: M.S. in Energy and Resources (also offered as M.A.), Energy and Resources Group (ERG), Rausser College of Natural Resources. Two-year, interdisciplinary, fall entry only, capstone project. About 20 of ~350 applicants admitted per year (MS + PhD combined). **Fit for you — strongest thematic fit in this tier:** your FYP (energy + water/irrigation trade-offs, GIS screening) maps almost exactly onto ERG's remit; floating PV → renewables integration and energy-systems analysis; ERG explicitly values field experience. **Funding:** possible, not guaranteed — master's students may in some cases be funded by faculty advisors (GSR/GSI posts, fee and tuition remission plus salary; Block Grants typically first semester only). **GRE:** not required — and not accepted ("GRE test scores are no longer required and will not be accepted or considered"). **Deadline:** December (1 Dec for Fall 2027). **Flag — address it head-on:** "ERG has a strong preference for applicants who have had at least two years of work experience in the field." You have internships and FYP fieldwork, not two years of work. Your SOP (Chapter 8) names this gap and answers it with fieldwork — do not hide it. Also: ERG says applicants "do not need, and are not encouraged, to solicit endorsement from a faculty member" — name professors in the application's faculty-interest section instead of cold-emailing for endorsement. **Page:** erg.berkeley.edu/graduate-programs/graduate-admissions

R3. Massachusetts Institute of Technology — SM in Technology and Policy (TPP)

Programme: Master of Science in Technology and Policy, Technology and Policy Program (TPP), Institute for Data, Systems, and Society (IDSS). Two-year, research-based, master's thesis; technical concentration (e.g. energy) + economics, politics, quantitative methods. Research assistantships across MIT labs and centres including the MIT Energy Initiative and CEEPR. **Why TPP, not MIT EECS:** the MIT EECS SM is **not open as a terminal MS to external international applicants** — it is required for PhD students and available to LGO students only. Do not apply to MIT EECS as a standalone MS. **Fit for you:** your FYP is fundamentally a trade-off problem (irrigation vs generation, economics of small hydro) — exactly TPP's technical-energy + economics + policy mix; Tarbela gives real-project power-sector grounding; floating PV/storage/smart grids map to a TPP energy concentration and thesis work on electricity systems and decarbonisation. **Funding — best in this tier, with a caveat:** research assistantships are overwhelmingly the most common support among TPP students — a typical 100% RA pays full tuition, a monthly stipend, and health insurance. Roughly half of each cohort has an RA by the fall start; bridge funding exists. But funding is not guaranteed in the offer letter — you must interview for RA posts after admission. **GRE:** optional. **English flag:** IELTS minimum **7.5**;

Duolingo 135; **TOEFL scores are not accepted** for TPP. Your IELTS planning must target 7.5, not 7.0, for this programme. **Deadline:** December (15 Dec). **Page:** oge.mit.edu/programs/technology-and-policy-program/

5.2 Target — 4 universities

T1. University of Michigan, Ann Arbor — MEng in Energy Systems Engineering

Programme: Master of Engineering in Energy Systems Engineering, College of Engineering (administered via Chemical Engineering). Note: at Ann Arbor this is an **MEng, not an MS** — "MS in Energy Systems Engineering" does not exist under that exact name. **Fit for you:** direct map — FYP (canal hydropower, physics-first ranking, irrigation-first, GIS) and energy-economics interest; core courses include Energy Generation and Storage Using Modern Materials, Sustainable Energy Systems, Energy Infrastructure Systems, plus energy-markets/policy and battery-systems options; floating PV → generation/storage core; Tarbela work → Power Systems Design and Operation. **Funding:** official page — "financial aid opportunities for graduate students are limited." Treat as largely self-funded. **GRE:** not required (but "strongly encouraged, especially for students with a GPA below 3.2" — your 3.46 clears that). **Deadline:** not stated on the admissions page — it says only "deadlines vary by semester and applicant type." Do not assume 15 January (that belongs to other Michigan Engineering master's programmes). Verify on the ESE deadlines page. **GPA minimum:** 3.0. **Page:** che.engin.umich.edu — ESE curriculum

Alternative at the same university (not the pick): MS in Electrical and Computer Engineering, Power & Energy area — stronger power-systems fit, but harder admit: GRE **required for international graduates**, no departmental financial aid for master's applicants, and "admission typically requires a minimum GPA of 3.5" — your 3.46 is just below that bar.

T2. University of Wisconsin–Madison — MS in Electrical and Computer Engineering (Research)

Programme: MS in ECE, **Research named option** (traditional two-year, thesis/research project) — the right on-campus route for you. Do not apply to the Power Engineering MS as the main route — it is online/off-campus for working professionals. **Fit for you:** the thesis route fits an applicant who wants research (FYP hydropower ranking, floating-PV field data, Tarbela 5th Extension). **Funding:** applicants are automatically put in the pool for TA funding; "funding decisions come directly from faculty members. However, funding is limited and never guaranteed. All admitted ECE MS-Research applicants should anticipate funding themselves." Plan to self-fund; a professor's RA is the upside. **GRE:** not required. **Deadline:** December — 1 December; no spring admission; minimum GPA 3.0. **Page:** guide.wisc.edu — ECE MS

T3. University of Colorado Boulder — MS in Electrical & Computer Engineering

Programme: MS in Electrical & Computer Engineering (traditional MS, thesis optional), Department of Electrical, Computer & Energy Engineering (ECEE). Focus areas include power electronics and Next Generation Power & Energy Systems. NREL

is next door. **Fit for you:** FYP, Tarbela hydro/switchyard experience, floating PV, renewable-integration and storage interests all map to the power/energy focus. **Funding — flag honestly:** the official Graduate Programs page says verbatim that master's degrees are "generally entirely unfunded" and "nearly all research assistantships (RA) go to fully fund our PhD students, and we do not use teaching assistantships (TA)." Traditional-MS students are eligible for an RA only "if positions become available". **Treat as self-funded. GRE:** not required ("not part of the review process"). **Deadline:** February — 15 February for all master's programmes. **Page:** colorado.edu/ecee — graduate programs

T4. North Carolina State University — MS in Electric Power Systems Engineering (MS-EPSE)

Programme: MS in Electric Power Systems Engineering (MS-EPSE), Department of Electrical and Computer Engineering, with the NSF FREEDM Systems Center. 30 credits, non-thesis, industry-sponsored capstone. **Strongest fit in this tier.** Also exists: MS in Electrical Engineering (thesis and non-thesis, Power Electronics and Power Systems area) — the thesis route if you want an RA. **Fit for you:** EPSE focuses on "modern electric power systems — including power generation, transmission, distribution, smart grid technologies, and renewable energy integration". Maps course-by-course: Power System Operation and Control, Smart Electric Power Distribution Systems, Renewable Electric Energy Systems, The Business of the Electric Utility Industry. FYP + Tarbela + floating PV + smart-grid interests all map. **Funding:** the ECE master's page states students are funded through fellowships, TA and RA appointments — but does not break this down by programme, and EPSE is a professional non-thesis programme, so funded places are likely scarcer than on thesis MS-EE. Possible but not assured — verify with the ECE Graduate Office or Prof. Baran (Chapter 6). **GRE:** not required (catalog entry: "Entrance Exam: None"). **Deadline:** January — 9 January (US and international). **Restriction:** the distance/online EPSE track is limited to people who work or reside in the United States — apply to the **on-campus** track. **Page:** ece.ncsu.edu — master's degrees · FREEDM: freedm.ncsu.edu/why-epse/

5.3 Likely — 3 universities

L1. Oregon State University — MS in Electrical and Computer Engineering

Programme: M.S. in Electrical and Computer Engineering, School of EECS, College of Engineering, Corvallis. Research area "Energy Systems" ("improving systems related to energy generation"). **Fit for you:** FYP → Energy Systems area; Tarbela → power/energy systems; floating PV → renewable energy. **Funding:** not stated on the official ECE programme page — do not assume funding. Treat as likely self-funded unless a professor offers an RA. **GRE: not stated on the official ECE programme page — unresolved.** Must be checked on the official apply page before applying; do not assume GRE-free. **Deadline:** January — 5 January (Fall 2027 priority consideration for MS). **Flag:** OSU's two best-known energy-systems professors both show "Not Accepting New Students" — Prof. Ted Brekken (energy systems / hydropower & storage) and Prof. Eduardo Cotilla-Sanchez (grid resilience, microgrids). The

programme fits, but outreach needs a different, currently-recruiting faculty member (check Annette von Jouanne). Do not cold-email Brekken or Cotilla-Sanchez expecting a place. **Page:** engineering.oregonstate.edu — ECE graduate

L2. University of Idaho — MS in Electrical Engineering

Programme: Electrical engineering, M.S. (thesis concentration only), Department of ECE, College of Engineering, Moscow, Idaho. 30 credits, STEM-designated. The official page calls Idaho "a leading hub for power engineering". **Fit for you — closest programme-level match in this tier:** FYP hydropower/power calculation and GIS screening; Tarbela switchyard/turbine/generator experience (directly relevant to power-systems coursework); floating PV and storage (power conversion, grid integration of variable renewables). Idaho is a hydropower state (Idaho National Laboratory, hydro-heavy grid). **Funding:** not promised on the degree page — "students are encouraged to secure a commitment from a faculty [member] for admission." Funding runs through a faculty advisor/RA, not a guaranteed departmental award. University-wide, graduate assistantships are open to international students; a 2025–26 TA stipend of $16,793 (9-month) is listed. Competitive, not guaranteed — apply by the priority date and secure a professor first. **GRE:** not required. **Deadline:** January (priority) / April (international final) — applications before 15 January receive full consideration for admission, financial support or assistantships; international final deadline 15 April for Fall. **Flag — degree background:** the catalogue requires a bachelor's in electrical engineering (GPA ≥ 3.0); non-EE applicants "may be admitted" only if they demonstrate proficiency in EE fundamentals (power area: circuits, power systems, machines). Your degree is Energy Engineering, not EE — foreground your power/electrical coursework and Tarbela electrical experience, and expect possible prerequisite conditions. **Page:** uidaho.edu — electrical engineering MS

L3. Montana State University — MS in Electrical Engineering

Programme: M.S. in Electrical Engineering (MSEE), Plan A (thesis) or Plan B (professional paper), 30 credits, Department of ECE, Bozeman, Montana. Graduate power courses include Power System Operation and Control, Power System Reliability, and Alternative Energy (wind, solar, microturbine, fuel cells, power electronic interfacing, energy storage). **Fit for you:** maps to FYP hydropower and power-system operation/reliability questions, Tarbela plant experience, and floating PV + storage. **Funding — best funding statement of the three Likely schools (official handbook):** "The majority of graduate students in our program receive financial support through teaching assistantships, research assistantships, and/or fellowships… All applicants to the M.S. or Ph.D. programs are considered for financial support without submitting any additional application." RAs generally pay a stipend, tuition, and fees; GTAs receive a monthly stipend and tuition waiver (international GTA applicants need TOEFL iBT ≥ 90 or IELTS ≥ 7.0). Funding is real but project-dependent — apply by the priority deadline and contact a professor. **GRE:** not required. **Deadline:** January (priority) / May (international final) — 15 January for full consideration with financial support; international students 15 May for Fall. **Page:** montana.edu — ECE graduate

Tier	University	Programme	GRE	Fall deadline (month)	MS funding reality
Reach	Stanford	MS Electrical Engineering	Not required	December (1 Dec)	Largely self-funded; Knight-Hennessy is the full-funding route
Reach	UC Berkeley	MS Energy and Resources (ERG)	Not required / not accepted	December (1 Dec)	GSR/GSI possible, not guaranteed
Reach	MIT	SM Technology and Policy	Optional	December (15 Dec)	RA presumptive model, not guaranteed in offer
Target	Michigan	MEng Energy Systems Engineering	Not required	Not stated — verify	Limited aid; largely self-funded
Target	Wisconsin–Madison	MS ECE — Research	Not required	December (1 Dec)	TA pool automatic; limited, never guaranteed
Target	Colorado Boulder	MS ECE (traditional)	Not required	February (15 Feb)	Generally entirely unfunded; RAs go to PhD
Target	NC State	MS Electric Power Systems Eng.	Not required	January (9 Jan)	Possible, not assured — verify
Likely	Oregon State	MS ECE	**Not stated — check**	January (5 Jan)	Not stated; likely self-funded unless a professor offers RA
Likely	Univ. of Idaho	MS Electrical Engineering	Not required	Jan (15 Jan priority; 15 Apr intl.)	Via faculty advisor/ RA; competitive
Likely	Montana State	MS Electrical Engineering	Not required	Jan (15 Jan priority; 15 May intl.)	Majority funded (RA/TA/fellowship); MS auto-considered

6. Professors — 8 verified contacts and how to write to them

For direct applications, the RA is funded by a professor's grant — so a specific, demonstrable match with a funded professor is the mechanism, not GPA alone. Contact professors **before** you apply (September–November 2027), concisely, with your CV attached. Direct faculty contact is explicitly encouraged by several departments (Michigan ME/ECE, Georgia Tech ECE) — it is not rude, and no opened page said it is penalised. One exception: Berkeley ERG says do not solicit endorsement — name professors in the application form instead.

Contact status matters: where a professor's faculty page shows a public email, it is given below. Where it does not, the booklet says so — never guess an email address. Six of the eight profiles below could not be opened directly (browser rate limits on the day of research) and are marked as such — re-open each faculty page before you write.

Professor	University / lab	Research (one line)	Verified paper / link	Contact
Prof. Ram Rajagopal	Stanford — Civil & Environmental Eng. and EE; Stanford Sustainable Systems Lab; Bits & Watts	Integration of renewables, smart distribution systems, demand-side data analytics	Wang et al. 2019, "Reliability Value of Distributed Solar plus Storage Systems…", IEEE Trans. Smart Grid. DOI: 10.1109/TSG.2018.2861227	No public email — conta department page. Profil profiles.stanford.edu/ra
Prof. Duncan Callaway	UC Berkeley — Chair, Energy and Resources Group; Lawrence Berkeley National Lab; EMAC group	Grid decarbonisation, power-system modelling, control and optimisation, renewables + storage integration	Chojkiewicz et al. 2026, "Leveraging existing grid infrastructure for rapid US renewable and storage deployment", PNAS. DOI: 10.1073/pnas.2603919123	No public email — conta department page. erg.b people/duncan-callaway no-endorsement rule ab
Prof. Christopher R. Knittel	MIT — Sloan; Director, MIT Center for Energy and Environmental Policy Research (CEEPR)	Energy economics, electricity pricing and policy, energy transitions	Jacobsen et al. 2020, "The Use of Regression Statistics to Analyze Imperfect Pricing Policies", J. Political Economy. DOI: 10.1086/705553	No public email — conta department page. A TPP listed as a graduate res at CEEPR working with P a TPP energy-economics realistic route.
Prof. Johanna L. Mathieu	Univ. of Michigan —	Modelling, estimation and	Granitsas et al. 2024, "Controlling air conditioners	**jlmath@umich.edu** (sl official umich.edu pages

	EECS; Director, Institute for Energy Solutions; Michigan Power and Energy Lab	control of electric loads and energy storage to integrate renewables	for frequency regulation", IEEE Trans. Smart Grid. DOI: 10.1109/TSG.2024.3513296	Power System Design a and Grid Integration of Energy Sources.
Prof. Bri-Mathias Hodge	Univ. of Colorado Boulder — ECEE (Chair); Renewable & Sustainable Energy Institute; formerly Chief Scientist, NREL	Modelling and simulation of power/energy systems; integrating wind and solar	Sajadi et al. 2022, "Synchronization in electric power networks… up to 100% inverter-based renewable generation", Nature Communications. DOI: 10.1038/s41467-022-30164-3	No public email — conta department page. As EC may have limited capac students — contact earl about MS (not PhD) sup
Prof. Mesut E. Baran	NC State — ECE; FREEDM Systems Center; programme lead for MS-EPSE	Distribution/transmission analysis and control, renewable integration, smart power distribution	Hu et al. 2021, "A Load Switching Group based Feeder-level Microgrid Energy Management Algorithm…", IEEE PES General Meeting. DOI: 10.1109/PESGM46819.2021.9638231	**baran@ncsu.edu** (FRE page). As EPSE program the right person to ask a campus EPSE admission capstone, and fellowshi availability for internati students.
Prof. Brian K. Johnson	Univ. of Idaho — ECE; SEL Endowed Chair in Power Engineering	Power system protection, energy storage systems, grid integration of inverter-based renewables	"How Inverter-Based Resources (IBRs) Affect Protection Relay Elements", UNIFI General Meeting 2025. DOI: 10.2172/3015386	**bjohnson@uidaho.edu** the official EE degree pa contact for Idaho — the itself tells thesis applica a faculty commitment, a the named EE contact.
Dr. Shamsunnahar Edib	Montana State — Assistant Professor, ECE (joined Spring 2025)	Cyber-physical power systems — resilience and restoration, smart-grid monitoring, optimisation, AI in energy systems	Edib et al. 2023, "Cyber Restoration of Power Systems…", IEEE Trans. Systems, Man, and Cybernetics: Systems. DOI: 10.1109/tsmc.2023.3258412	**shamsunnahar.edib@** (official MSU ECE Gradu Handbook). Flag: her pu students specified PhD s email to ask explicitly a supervision before assu

Gap to close: no Wisconsin professor is included above. Before you apply to Wisconsin–Madison, find one WEMPEC (Wisconsin Electric Machines and Power Electronics Consortium) or power-engineering faculty member whose research matches yours, from the official ECE pages.

6.1 How to write to a professor — the method and the email

Steps: (1) read one of their papers (the DOI in the table) and find one specific link to your work; (2) write one short email — never more than 200 words; (3) attach only your CV in the first email; (4) if no reply in two weeks, send one polite reminder (your Pakistan Scholarships League documents include a reminder-email body — use it); (5) never email the same professor twice in one week, and never email two professors in the same department on the same day with the same text. **What to ask:** whether they are taking MS thesis students for Fall 2028, and whether an RA might be available — not “please fund me”.

First email to a professor (copy, fill the brackets, keep under 200 words):

Subject: Prospective MS student (Fall 2028) — hydropower / power systems — [Your Name]

Dear Professor [Name],

I am Muhammad Ismail, a final-year BS Energy Engineering student at UET Peshawar, Pakistan (GPA 3.46, graduating October 2027). My final-year project assesses low-head hydropower from KP irrigation canals — twelve sites ranked by measured head and seasonal flow, with irrigation as a first constraint — and I have worked at WAPDA’s Tarbela station and on the Tarbela 5th Extension project.

I read your paper [title, year] on [one-line link to your work, e.g. “inverter-based resource integration”]. My question is whether you are taking MS thesis students for Fall 2028, and whether a research assistantship might be available in your group. I have attached my CV (2 pages). I would be grateful for any guidance, including if another colleague in your department would be a better fit.

Thank you for your time.

Kind regards,
Muhammad Ismail
[email] · [phone] · [GitHub/CanalPower link]

What makes a professor reply: a specific paper cited, a specific skill offered (field data, GIS screening, MATLAB/Python modelling, plant experience), and a short email. What makes them delete it: a long life story, a generic “I am very passionate”, an attachment list of ten files, or a demand for funding in the first line. You already have professor-outreach documents from the Pakistan Scholarships League group (11 documents, delivered 2 November 2026) — use their templates alongside this chapter, not instead of it.

7. The projects that make you competitive

Same seven projects as the Erasmus booklet, in the same risk order — but read through a US lens: a thesis-based MS and a professor’s RA are won by research evidence, so every project below ends with the line it gives your SOP and the line it

gives a professor's email. **Nothing new starts before the FYP presentation on 6 November 2026.**

Project	Why it matters for the USA	Dated steps (summary)	Done test
1. FYP — low-head hydropower, KP canals (Nov 2026 – mid 2027)	The centre of every SOP and your RA pitch. A finished thesis with a defended ranking is research experience a professor can fund.	6 Nov 2026 presentation → Nov–Dec: close Warsak data request → Dec–Feb: measure head/discharge → Feb–Apr: physics-first ranking + LCOE/NPV/IRR → Apr–Jun: thesis + defence	Thesis PDF + one-page ranked site table with sourced head/flow
2. CanalPower PK (Nov 2026 – mid 2027)	Open-source proof of system thinking and Python progress; a link a professor can open. Start after 6 Nov, 45–60 min/day, ~8 weeks.	Week 1: walkthrough + tests pass → Weeks 2–3: io.py + test → Weeks 4–5: Streamlit app → Weeks 6–8: sensitivity, QGIS export, README → mid-2027: public repo	Public GitHub repo + running demo screenshot
3. Energiscope honesty repair (November 2026 — do first)	The fastest credibility win; the only project that can actively harm an application if left as it is (Chapter 1).	Upload 6 P0 files from Energiscope_P0_Fixes.zip → verify live site → Step B: remove hardcoded admin password, fix sample links, one canonical dataset, Methods page → 2027: replace sample data with sourced NEPRA data	Live site shows honesty banner + Methods page
4. HOMER Pro — Gulabad hybrid (Dec 2026 – Mar 2027)	Your only techno-economic modelling proof; maps to energy-systems and energy-economics programmes and to RA work on hybrid systems.	Finish honest load profile + NASA POWER CSV first → start 21-day trial only then (Windows lab/borrowed PC/VM; never cracked software) → PV–hydro–battery vs PV–diesel–battery → record NPC, cost of energy, renewable fraction + one sensitivity → 3–4 page report	Report PDF with HOMER screenshots and the four numbers
5. Floating-PV manuscript pack (Dec 2026 – Apr 2027)	Your research experience — the best evidence for an RA. 120 weekdays of field data is a line	Audit workbook (one headline table, three figures with units) → manuscript skeleton in the supervisor's loop → stays "in preparation" until an actual submission exists	Manuscript-pack folder you can show a professor

	professors remember.		
6. Skills proof — Python, MATLAB, GIS (Nov 2026 – Sep 2027)	RA work is modelling work. "Developing" must become visibly "improving" by application time.	Python: finish Lesson 01 independent task (Site F; CF 0.45; head for 500 kW) → MATLAB: settle Simscape licence, PV Array model or single-diode fallback, I–V/P–V curves → GIS: finish corridor clip (corridor_mask → dem_clipped.tif), state the 30 m DEM limit on the map	One folder per skill with a runnable file + screenshot
7. Tarbela records + coursework evidence (one weekend, early 2027)	Plant experience is what separates you from classroom-only applicants — but only if it is documented.	One-page summary per placement (operated vs observed, dates, supervisor name, certificate requested in writing) → six one-page coursework evidence sheets ("inputs not retained — method retained" where numbers are gone)	Two placement summaries + six evidence sheets

8. SOPs — the US method, then three full drafts

US statements of purpose differ from Erasmus motivation letters in two ways: they are more research-focused (what you have done, what you want to research, with whom), and Fulbright's two essays are different documents again (Study Objectives = academic plan; Personal Statement = who you are and why you will return and serve). Write each one for its exact audience.

8.1 The method — five paragraphs for a university SOP

Para	Job	What goes in (yours)	What never goes in
1. Hook	One real moment	Charsadda canals; water doing farm work and nothing else	"Since my childhood I have always been passionate…" with no moment attached
2. Research proof	What you did, with numbers	FYP (12 sites, physics-first ranking, irrigation veto); floating PV (120 weekdays); Tarbela plant work	Claimed results, "validated" models, invented site power figures
3. Building + skills	What you made, what you can do	One project (Energiscope / digital twin / CanalPower PK) + Python, MATLAB, GIS status honestly stated	A list of every project; "proficient" for a skill still in progress

4. Why this programme + professor	The only fully rewritten paragraph	Named professor, named paper, named lab, named courses — and the research question you want to work on	Flattery, or a programme name with no professor or course behind it
5. Future	What you do with the degree	Return to Pakistan's energy sector; thesis → practice; Fulbright version adds the 2-year service commitment explicitly	Vague "contribute to society" with no job, sector, or place

Length: 500–1,000 words unless a programme states otherwise — check each programme's page; Fulbright essays run to about 800 words each per USEFP's own instructions (quote those, not guidebooks, where the two conflict). **Rule:** these drafts are starting points in your voice, built only from verified facts. Rewrite each one in your own words, read it aloud, and get it checked by a scholarship holder or an EducationUSA adviser (Chapter 14) before you submit. Never submit a draft word-for-word.

8.2 SOP draft 1 — Fulbright Study Objectives essay

I grew up in Charsadda, in Khyber Pakhtunkhwa, where irrigation canals run past communities that the electricity grid serves poorly. Every day I watched water do one job — carry water to farms — and nothing else. At each fall and regulator, the water dropped through a height and its energy was simply lost. That sight is the reason I chose Energy Engineering, and it is the reason for my final-year project.

At UET Peshawar I am assessing low-head hydropower from the KP irrigation network: twelve canal sites, ranked by a physics-first method — power from measured head and seasonal flow, $P = \rho g Q H \eta$ — with every site screened first against irrigation. A site that harms irrigation timing, or works only when irrigation does not need the water, is screened out. Alongside the project I have worked inside real hydropower: four weeks at WAPDA's Tarbela station, on electrical maintenance, switchyard, turbines and generators, and since September 2026 as a project intern on the Tarbela 5th Extension project. I have also contributed to UET floating-photovoltaic research, collecting 120 weekdays of field data to benchmark floating against ground-mounted PV.

For my master's I want to study decentralised energy systems: resource assessment, system modelling, storage, and grid integration of small renewable generation. My objective is a thesis-based MS in which I can extend my FYP method — physics-first ranking with an irrigation constraint — into a general framework for small, distributed generation on existing water networks, and test it against the grid-integration question my FYP leaves open: how do small sites connect to a grid, stay stable, and deliver power where it is needed? I am particularly interested in programmes with a power-systems and renewables-integration focus, and in working with a faculty advisor whose research covers distributed generation and storage.

I build the tools I use. I built Energiscope, a public dashboard for generation data, and I am now rebuilding its dataset honestly after learning that its first version used

sample data. I built a working digital twin of my canal network and I am building CanalPower PK, an open screening tool for KP canals. Each project taught me that a good engineering idea is worthless without honest data and a real constraint — a lesson I will carry into graduate research.

After the master's I will return to Pakistan, as the Fulbright programme requires, and serve for at least two years in hydropower and renewables work — in project engineering or research — applying the same method to real sites: physics first, irrigation first, economics included. Pakistan spends about USD 14.5 billion a year on energy imports while canal hydropower already works here — the Machai plant in Mardan generates about 14 million units a year from 2.6 MW on a canal. Closing that gap is the work I want to do, and a US master's is how I will learn to do it properly.

Before you submit this one: check USEFP's own essay instructions when the 2028 call opens (word limit, prompts) and reshape the draft to them. Update the FYP sentence when the ranking is actually run. The return commitment in the last paragraph is not optional — Fulbright selects people who mean it, and the contract binds you to it.

8.3 SOP draft 2 — University SOP (power-systems MS: NC State / Wisconsin / Idaho / Montana State)

Dear Admissions Committee,

I grew up in Charsadda, in Khyber Pakhtunkhwa, where irrigation canals run past communities that the electricity grid serves poorly. I chose Energy Engineering because I wanted to understand both ends of that problem: the generation that does not exist yet, and the grid that must carry it when it does.

At UET Peshawar my final-year project assesses low-head hydropower from the KP irrigation network — twelve canal sites, ranked by measured head and seasonal flow, with irrigation as a hard constraint. But the experience that most shaped my choice happened at Tarbela. In four weeks at WAPDA's Tarbela station I worked on electrical maintenance, switchyard operations, turbines, generators and high-voltage systems, and saw real-time plant load coordination. Since September 2026 I have continued as a project intern on the Tarbela 5th Extension project, with hands-on electrical and mechanical work including installation and rewiring. I have seen, at plant scale, the system my small canal sites would one day feed into — and I have seen how much of the engineering is electrical, not civil.

My research preparation is practical. I contributed to UET floating-photovoltaic research, collecting 120 weekdays of field data and building the comparative workbook for floating versus ground-mounted performance. My coursework includes Power Systems, Control Systems, power system analysis and protection, and basic electrical circuits, with laboratory models including a DC-DC boost converter for solar PV output regulation in MATLAB. I am building Python, MATLAB and GIS skills through project work — a screening tool, a PV model, and a QGIS corridor map — and I am honest about where they stand: developing, not yet proficient.

I am applying to your programme because of its power-systems focus — [named courses, e.g. Power System Operation and Control; Smart Electric Power Distribution Systems; Renewable Electric Energy Systems] — and because I want to work with

Professor [name] on [their research area, citing one paper]. My research question is the one my FYP leaves open: how do small, distributed hydropower sites connect to a grid, stay stable, and deliver power where it is needed? I am applying for a thesis-based MS because I want to answer that question with research, and because I am seeking a research assistantship in your group.

After the master's I will return to Pakistan's energy sector, to work on the integration of distributed renewables into grids that were not designed for them. The generation matters, but the connection is what makes it useful — and that is the engineering I want to master.

Yours sincerely,
Muhammad Ismail

Before you submit this one: fill in the bracketed courses and professor for each university — the NC State version names EPSE courses and Prof. Baran; the Idaho version names Prof. Johnson and the power/protection focus; the Montana State version names Dr. Edib and resilience. A letter with brackets still in it, or with another university's professor named, is a rejection. For Wisconsin, find the WEMPEC/power professor first (Chapter 6 gap).

8.4 SOP draft 3 — Policy/energy SOP (MIT TPP / Berkeley ERG)

Dear Admissions Committee,

I grew up in Charsadda, in Khyber Pakhtunkhwa, where irrigation canals run past communities that the electricity grid serves poorly. My final-year project at UET Peshawar asks whether the unused head at those canal falls can become electricity. But the longer I worked on it, the clearer a second question became: even if the engineering works, who pays for it, how is it priced, and does the money make sense? A site that generates power nobody can afford to build is not a solution. That question — technical, economic and institutional at once — is why I am applying to your programme.

My project assesses twelve canal sites in the KP irrigation network, ranked by measured head and seasonal flow, with irrigation as a hard constraint. Its method is physics-first, but its final chapter is economics: my thesis will compute the levelised cost of electricity, net present value, internal rate of return and payback period for each site, because a ranking by power alone is not a feasibility study. I have also worked inside real hydropower at Tarbela, and contributed to floating-PV research at UET, so I have seen both the plant and the spreadsheet side of energy decisions.

The economics of energy in Pakistan is not abstract to me. The country spends about USD 14.5 billion a year on energy imports — nearly a fifth of its foreign-exchange reserves — while most generation still comes from fossil fuels. At the same time, canal hydropower already works here: the Machai plant in Mardan generates about 14 million units a year from 2.6 MW on a canal. The gap between those two facts is an economics and policy problem as much as an engineering one, and it is the problem I want to spend my career on.

[MIT TPP version:] I want TPP's combination — a technical energy concentration with economics, politics and quantitative methods — and the chance to do thesis research with the MIT Energy Initiative or CEEPR, where electricity pricing and policy are

studied as seriously as the engineering. [Berkeley ERG version:] I want ERG's interdisciplinary training — energy systems analysis with a water and development dimension — and I want to answer, honestly, the work-experience question: I do not have two years of employment, but I have two years of fieldwork — canal surveys, irrigation-official negotiations, plant work at Tarbela, and 120 weekdays of floating-PV field data. That fieldwork is the experience I bring.

After the master's I will return to Pakistan to work where engineering and policy meet — in energy project development, research, or policy analysis — as the person in the room who understands both the turbine and the tariff.

Yours sincerely,
Muhammad Ismail

Before you submit this one: use only one bracketed version — TPP or ERG — and delete the other. For TPP, your IELTS must be 7.5 (TOEFL is not accepted). For ERG, do not solicit a faculty endorsement — name professors in the application's faculty-interest section instead. When your thesis economics are computed, replace "my thesis will compute…" with the real result for your best site.

> **Common SOP mistakes — check every draft against this list.** (1) Copying a template or another scholar's letter. (2) Adjectives without evidence. (3) Claimed results you do not have ("validated model", "published paper", "10+ years of real data"). (4) Writing about the university's prestige instead of its professors, labs and courses. (5) One SOP for all universities with only the name changed. (6) A Fulbright essay that does not mention returning to Pakistan. (7) No proofreading — get a scholarship holder and an EducationUSA adviser to read each essay (Chapter 14).

9. LORs — who to ask, when, and how Fulbright references differ

US universities usually ask for three letters of recommendation; Fulbright asks for three reference letters. Same three referees, but the two routes need slightly different letters — and Fulbright's referees submit through USEFP's system, while universities use their own portals (each portal emails the referee a link). Plan for three referees, so you are covered either way.

9.1 Who exactly to ask — your three referees

#	Referee (get the exact name + title)	What they can honestly say	Serves best
1 — Academic, strongest	**Prof. Dr. Farid Ullah Khan**, your FYP supervisor, UET Peshawar	Your FYP: problem choice, method, fieldwork, honesty about limits, ranking work, thesis quality. The	All universities + Fulbright. Ask him first.

		only person who can speak about your core project in detail.	
2 — Professional / technical	**Your supervisor at the Tarbela 5th Extension project (PCCCL)** — get their full name, title and work email before you ask	Hands-on plant work: installation, rewiring, electrical and mechanical components, reliability, teamwork on a live site.	Power-systems programmes (NC State, Idaho, Montana State, Wisconsin); Fulbright (practical leadership).
3 — Research	**Your floating-PV research lead at UET** — get their full name, title and email; if they cannot write, use the UET professor who taught Wind & Hydropower or Power Systems and graded your project work	Field data collection (120 weekdays), workbook discipline, comparative analysis, research conduct.	Thesis-MS and RA pitches (Berkeley ERG, MIT TPP, Colorado); Fulbright (research ability).

Never ask: family, friends, politicians, or a senior professor who does not know your work. **Waive your right to read the letters** when a portal asks — confidential letters are taken more seriously, and some programmes require the waiver. **Never write your own letter and ask a referee to sign it.** Give them facts and reminders (Section 9.3); the words must be theirs.

9.2 When — the dated steps (note: Fulbright is earlier)

1. **January–February 2027 — build the relationship.** Make sure referees 2 and 3 have seen your recent work before you ask: share the HOMER report, the floating-PV figures, or the CanalPower demo.
2. **February 2027 — ask in person for Fulbright.** Visit Prof. Dr. Farid Ullah Khan in his office. Say: “I am applying for the Fulbright Foreign Student Program, deadline 1 April. Would you be willing to write a strong reference for me?” Same for referees 2 and 3, in person if possible. Follow up the same day with the email in 9.4.
3. **Within 24 hours of the yes — send the recommender pack (9.3).** One email, one folder: CV/résumé, your essay drafts, transcript, a one-page bullet list of what they saw you do, deadlines, and how the letter is submitted.
4. **Two weeks before the Fulbright deadline — gentle reminder.** One short email. Thank them, restate the date, offer to resend the link.
5. **September–October 2027 — ask again for direct applications.** Your referees already wrote once; now they adapt the letter per university. Send each one the university list, the SOP for that university, and the portal links as they arrive. Remind them two weeks before the earliest university deadline.
6. **After submission — thank you + outcome.** A short thank-you within a week of the last deadline, and a message when results arrive, whatever the result.

Give referees a personal deadline two weeks before each real deadline — and tell them honestly that it is your personal deadline, so a delay cannot sink the application.

9.3 The recommender pack — what to send each referee

- Your current résumé (Chapter 10) and unofficial transcript.
- Your essay/SOP drafts (Chapter 8) for the route they are supporting, so the letter and the essay tell the same story.
- A one-page bullet list: “What you saw me do” — 5–7 bullets, each a specific moment with a date. Write a different list per referee, from their experience of you.
- One paragraph per programme: what it selects for, and what you would like the letter to stress (matrix below).
- Deadlines table (programme, personal deadline, real deadline, how to submit).

Referee	Fulbright	Power-systems MS (NC State / Idaho / Montana)	ERG / TPP
Prof. Khan (FYP)	Method, fieldwork, leadership, commitment to return and serve Pakistan	Systems thinking; physics-first ranking; irrigation constraint	Trade-off thinking; economics thread; honest limits
Tarbela supervisor	Reliability, teamwork, practical leadership	Switchyard, generators, maintenance, live-site work	Plant-scale grounding
Floating-PV lead	Research ability, persistence (120 weekdays)	Measurement discipline	Water-surface energy; comparative analysis

9.4 The ask email + one draft letter

Ask email (Fulbright version — send within 24 hours of the in-person yes):

Subject: Reference for Fulbright application — Muhammad Ismail (deadline 1 April 2027)

Dear Professor [Name],

Thank you for agreeing to write a reference for my Fulbright Foreign Student Program application (USEFP), for a master’s in the United States starting Fall 2028. The deadline is 1 April 2027.

Attached is a short pack: my résumé, my draft essays, my transcript, a one-page list of the work you supervised, and the submission instructions. My personal deadline for the reference is [date — two weeks before 1 April], so that a delay cannot affect the application.

What would help most is a reference that speaks to [one line per referee, e.g. “the FYP method and fieldwork you supervised, and my commitment to return and serve

Pakistan”]. Please let me know if you would like anything else, or if a short meeting would help.

Thank you very much for your support.

Kind regards,
Muhammad Ismail
BS Energy Engineering, UET Peshawar
[phone] · [email]

Draft letter — academic referee (for Prof. Dr. Farid Ullah Khan to adapt in his own words; send only if he asks for a draft):

To the Selection Committee,

I am Professor Dr. Farid Ullah Khan, [title/department], University of Engineering and Technology Peshawar, and I have supervised Muhammad Ismail’s final-year project since [month/year]. I write in support of his application.

Mr. Ismail’s project assesses low-head hydropower from the KP irrigation network — twelve canal sites, ranked by measured head and seasonal flow, with irrigation as a first constraint. What distinguishes his work is method: he insisted on a physics-first ranking, with machine learning only as a comparative check, and he refused to state power figures before head was measured. In my experience that discipline is rare in an undergraduate, and it is the reason I trust his results.

He carried out field reconnaissance at Gulabad–Sardheri and Shabqadar, engaged irrigation officials at Warsak, and built a GIS-based site screening workflow. He also built an open screening tool, CanalPower PK, and a working digital twin of the canal network. [Referee: add one honest sentence about his presentation or thesis work, and one specific example you observed — a decision, a correction, a field problem he solved.]

Academically he is a strong student (GPA 3.46) whose best work is practical: he learns fastest when a real site or a real dataset is in front of him. I have also seen him lead his group of three through a difficult, data-scarce project without cutting corners. He has told me plainly that he intends to return to Pakistan and work in hydropower and renewables, and I believe him — his project was chosen for Pakistan’s needs, not for a transcript.

I recommend Mr. Ismail strongly. He has the technical grounding, the field experience, and the honesty about evidence that graduate study requires. Please contact me if you need anything further.

Yours sincerely,
Prof. Dr. Farid Ullah Khan
[Title, Department of Energy Engineering, UET Peshawar]
[Email] · [Phone]

10. Your US résumé — line by line (not Europass)

US universities want a résumé, not a Europass CV: 1–2 pages, reverse-chronological, action verbs, numbers where you have them. Your refined 2-page CV is the base — these are the changes for the USA.

Section	Keep / change for the USA
Header	Name, email, phone (+92 314 9489386), LinkedIn, GitHub. No photo, no date of birth, no nationality line on the résumé itself (that goes in the application form).
Education	BS Energy Engineering, UET Peshawar, Oct 2023 – Oct 2027 (expected), GPA 3.46/4.0. Relevant coursework only (the eight courses in Chapter 1).
Research experience	FYP first (12 sites, physics-first ranking, irrigation-first, GIS screening, digital twin link). Floating PV second (120 weekdays field data, comparative workbook, “manuscript in preparation”).
Professional experience	Tarbela 5th Extension (PCCCL, Sep 2026, hybrid, installation and rewiring) and WAPDA Tarbela (4 weeks, switchyard/turbines/ generators). Separate entries, with dates.
Projects	Energiscope (after the P0 fix: “sample dataset 2020–2025; methods page in progress” + live link + GitHub), CanalPower PK (repo link once public), HOMER Gulabad (in progress, with status).
Skills	Honest levels: MATLAB (working proficiency); Python (developing); HOMER Pro (in progress); AutoCAD, Energy3D, Helioscope, Revit; NASA POWER, Global Solar Atlas, Global Wind Atlas; advanced Excel. Never “proficient” for a skill still in progress.
Languages / certificates / leadership	Pashto (native), Urdu (fluent), English (professional working proficiency; test scores added when they exist). Seven certificates with providers, at the bottom. Leadership in three lines (GIZ, AEE, SESA).
Delete	Any claimed publication; any claimed job; any skill level you cannot defend in an interview; the “10+ years real NEPRA data” line (until fixed); Europass formatting.

Steps: (1) rebuild the refined CV as a US résumé (same facts, new format); (2) keep it to two pages maximum — one page is fine if it fits, two is the limit; (3) get it read by an EducationUSA adviser and one scholarship holder; (4) save as Resume_Ismail.pdf and use the same file for Fulbright and for direct applications — only the SOP changes per university.

11. Tests — GRE (you must take it) and TOEFL/IELTS

11.1 GRE General Test — mandatory for Fulbright, useful for direct applications

Format (shorter test, in force since September 2023): Analytical Writing — 1 essay ("Analyze an Issue"), 30 minutes; Verbal Reasoning — 2 sections, 27 questions, 41 minutes; Quantitative Reasoning — 2 sections, 27 questions, 47 minutes. Total: 54 questions + 1 essay, about 1 hour 58 minutes. **Scores:** Verbal 130–170, Quant 130–170, Analytical Writing 0–6. ETS reports no combined total — "300" or "320" is an informal sum of Verbal + Quant, not an ETS score. **Validity:** 5 years. **Cost:** $249 (global rate; confirm at checkout — no Pakistan-specific fee was verified). Rescheduling $55; additional score reports $40 each. **ScoreSelect:** on retaking, choose which scores from the last 5 years to send.

Your targets: Quant **165+**, Verbal **155+**, Analytical Writing **4.0**. Fulbright's floor is 145 per section, but the floor is not the target — a 300+ combined score also makes you eligible to request the GRE fee reimbursement (public university, engineering, first-come-first-served, up to 30 requests). A strong Quant score also helps where a direct programme treats the GRE as optional.

Dated plan:

1. **December 2026 – January 2027:** diagnostic — one full official ETS practice test (free PowerPrep) under timed conditions. Score all three sections. This is your real starting point; do not guess it.
2. **January – March 2027:** 60–90 minutes a day. Quant: ETS Official Guide + Manhattan 5 lb. (the standard books) — your engineering maths is the strength; the risk is careless errors under time pressure. Verbal: vocabulary in context (not word lists alone) + reading comprehension practice. Writing: one "Analyze an Issue" essay a week, against the ETS scoring guide.
3. **By mid-March 2027:** take the GRE (test centre or at-home — check availability in Pakistan on the ETS site). Leave time for one retake before 1 April if the first score is below target.
4. **Send scores** to USEFP code **9388** and IIE code **2326**. Upload the score with the Fulbright application; unofficial scores + proof of registration are accepted if the official report is late.

11.2 TOEFL vs IELTS — which one, and when

The pattern: TOEFL is the traditional default and is accepted everywhere; IELTS Academic is now widely accepted in the USA. Typical engineering-MS minimums: **TOEFL iBT 85–100 / IELTS 6.5–7.0**, with 90/7.0 the most common published minimum and 100 the de facto competitive level. **Your targets: TOEFL iBT 100, or IELTS 7.0 (aim 7.5 for MIT TPP, which does not accept TOEFL at all).**

Which test to take: take **TOEFL** if you are applying for Fulbright (only TOEFL is taken from shortlisted candidates; IELTS is not mentioned on USEFP pages) or to

programmes that publish a TOEFL minimum. Take **IELTS** as well only if a programme on your final list requires it — MIT TPP is the one on this list (IELTS 7.5 minimum, TOEFL not accepted). Do not pay for both tests before you need them.

Timing: Fulbright — TOEFL only after shortlisting, so do not pay for it before April 2027. Direct applications — take the test in mid-2027 (June–September) so scores are fresh for December–January deadlines. Both tests are valid 2 years. **Note the TOEFL rescoring:** from 21 January 2026, TOEFL iBT scores are reported on a 1–6 scale (CEFR-aligned); during a two-year transition to January 2028, reports also show a comparable 0–120 overall score (old 100 ≈ 5, old 90 ≈ 4.5). Always check whether a programme's published minimum is on the old or the new scale.

Speaking matters twice: admission minimums are lower than the speaking standard required to hold a TA. At several universities, TA oral-English certification needs a TOEFL Speaking score well above the admission minimum (for example, Purdue's GTA standard is Speaking ≥27 on the old scale) or an IELTS Speaking score of 8.0, plus a university oral test. If you hope for a TA, give the Speaking section disproportionate preparation. **Costs (verify at booking):** TOEFL Pakistan fee was not verifiable on the ETS page (secondary sources report ~$230); IELTS Pakistan fee was not verifiable on an official page (secondary guides converge on about PKR 75,600). Confirm both on the official ETS / British Council Pakistan / IDP pages before paying.

12. Money — what it really costs, proof of funds, the I-20, and the F-1 visa

All figures below are 2025–26/2026–27 rates. No university has published 2028–29 tuition — every figure will rise before Fall 2028. Re-check each university's bursar page in 2027.

12.1 The honest budget (before any funding)

Stage	Realistic amount
Applications (6 public universities)	**$450–$630** ($75–$114 each, non-refundable; usually no fee waiver for international applicants)
GRE	**$249** (reimbursement request possible at 300+ — Chapter 4.4)
English test	TOEFL ~$230 (unverified — confirm at booking); IELTS ~PKR 75,600 (unverified — confirm on British Council/IDP Pakistan)
SEVIS I-901 fee + F-1 visa (MRV) fee	**$535** ($350 SEVIS + $185 MRV), plus any reciprocity fee — not verified for Pakistan; plus a possible $250 Visa Integrity Fee if it is activated by 2027–28 (it did not appear on the official State Department fee schedule opened 5 Nov 2026 — budget it as a risk, re-check at the time)
First-year tuition + fees (no funding)	**~$29,000–$44,000** per year (public universities, international rate; engineering differential fees extra)

First-year total cost incl. living + insurance (no funding)	**~$46,000–$62,000** per year
Health insurance	**~$2,000–$2,500** per year (effectively mandatory, usually billed automatically; sometimes waived with a qualifying assistantship)
First year with a **full RA/TA**	Tuition largely covered (tuition waiver + stipend, e.g. ~$2,300/month master's GRA at Georgia Tech); out-of-pocket mainly the $535 government fees, application fees, insurance if not covered, travel, and initial living costs until the stipend starts

12.2 Proof of funds and the I-20 — the step most applicants misunderstand

After admission, the university's Designated School Official must collect evidence of financial ability **before issuing the Form I-20**. F-1 rules require proof for **one year** of study. Documents that count: family bank statements (savings/checking, current balance), fixed deposits only if the bank notes funds can be withdrawn at any time without penalty, approved loan documents/sanction letters, scholarship award letters, and graduate assistantship offer letters — all dated less than 6 months old, representing liquid funds. A signed Letter of Financial Support is required if any document is in someone else's name (for example a parent). **Not acceptable:** stocks/bonds, real-estate holdings, retirement funds, life-insurance policies, unapproved loan documents, pay slips alone, documents older than 6 months. **The key arithmetic:** a scholarship or assistantship reduces the amount you must prove — a full RA/TA with tuition remission + stipend can reduce personal proof-of-funds to near zero (the I-20 lists the assistantship as the funding source); a partial award reduces it only by the documented amount. The award letter must exist before the I-20 is issued.

12.3 The F-1 visa — steps after the I-20

1. **Receive the I-20** from the university. Check every detail (name spelling, programme, dates, funding) before you pay anything.
2. **Pay the SEVIS I-901 fee ($350)** online at fmjfee.com. Keep the printed confirmation — the SEVIS ID must match the I-20.
3. **Complete the DS-160** visa application online and **pay the MRV fee ($185)**.
4. **Book the visa interview** at the US Embassy in Islamabad (or the Consulate in Karachi, as applicable). Book early — appointment waits vary.
5. **Attend the interview** with: passport, I-20, SEVIS payment confirmation, DS-160 confirmation, admission letter, financial documents, academic transcripts, test scores, and proof of ties to Pakistan (family, property, job prospects, your FYP and return plan). The interview is short — often a few minutes — and the questions are usually: why this university, why this programme, who is funding you, what will you do after the degree. Answer in your own words, consistently with your SOP. See the verified Islamabad walkthrough video in Chapter 15.

6. **After approval:** the visa is placed in your passport; you may normally enter the US up to 30 days before the programme start date. Do not book a non-refundable flight before the visa is in hand.

Fulbright grantees travel on a J-1 visa arranged with USEFP's help — the steps above are for direct-admission F-1 applicants. Do not mix the two visa routes.

13. Month-by-month timeline: November 2026 to September 2028

The USA timeline is front-loaded: Fulbright's deadline (~1 April 2027) comes before you graduate, while direct applications close December 2027 – February 2028. Miss April 2027 and Fulbright is gone for this cycle.

When	What to do (steps, not wishes)	Done test
Nov 2026	FYP second presentation (6 Nov). Energiscope P0 upload (1 hour). Start CanalPower PK Week 1. Email USEFP (info@usefp.org) with the parallel-applications question (Chapter 3) and keep the reply.	Presentation delivered; live site shows honesty banner; USEFP reply filed
Dec 2026	Check the USEFP Fulbright page for the 2028 call. GRE diagnostic (one full ETS PowerPrep, timed). FYP measurements begin. CanalPower Weeks 2–3.	Diagnostic GRE scores known
Jan–Feb 2027	GRE preparation, 60–90 min/day. Ask referees in person for Fulbright (Feb). Send recommender packs within 24 hours. FYP measurements continue; HOMER data ready.	3 referees confirmed for Fulbright
Feb–Mar 2027	Write the two Fulbright essays (Chapter 8.2 + Personal Statement). GRE — take it by mid-March; retake once before 1 April if below target. Remind referees two weeks before the deadline.	GRE score ≥300 combined (target Quant 165+, Verbal 155+); essays drafted and read by a holder/adviser
~1 Apr 2027	**Submit the Fulbright application.** Check the portal the next day: form complete, 3 references received, GRE report uploaded, transcripts attached.	Application "complete" in the USEFP portal
Apr–Jun 2027	FYP ranking + economics; HOMER report; CanalPower Weeks 6–8; floating-PV pack; MATLAB PV model. TOEFL preparation begins (do not pay for the test yet).	Ranked site table; HOMER report; CanalPower repo public
Jun–Aug 2027	Fulbright interviews (if shortlisted). Thesis writing. TOEFL test (mid-2027) for direct applications. IELTS as well only if MIT TPP stays on your final list (needs 7.5).	Thesis draft to supervisor; TOEFL score in hand
Sep 2027	Fulbright results (Principal/Alternate, September). Thesis defence (date TBC). **Freeze the direct-application list (6–8 universities):** re-check every programme's GRE rule, English minimum (old vs new	Final list frozen; tracker filled; résumé final

	TOEFL scale), deadline, fee and funding page. Fill the tracker (Chapter 16). Ask referees again for direct applications.	
Oct–Nov 2027	Write SOPs university by university (Chapter 8). Contact professors (Chapter 6) — September–November, before applications. Get each SOP read by a holder/adviser.	6–8 SOPs drafted and reviewed; professor emails sent
Dec 2027 – Feb 2028	Submit direct applications, earliest deadline first (Stanford/Berkeley/Wisconsin 1 Dec; NC State 9 Jan; Idaho/Montana State 15 Jan priority; Colorado 15 Feb). Check each portal the day after: documents received, LORs received. Keep confirmations.	All applications "complete" in every portal
Feb–May 2028	Results arrive. Fulbright placement continues (if selected) until April. Compare offers with funding letters — an admission without funding is not a funded place. Decide by each offer's deadline.	Decision made; acceptance + deposit sent as instructed
Jun–Aug 2028	I-20 → SEVIS $350 → DS-160 → MRV $185 → visa interview (Chapter 12). Housing, insurance, travel. HEC attestation only if asked. Tell your referees the outcome.	Visa in hand; flight booked
Sep 2028	Start the master's.	—

If you fall behind: protect, in this order — (1) the GRE before 1 April 2027 (Fulbright depends on it), (2) the two Fulbright essays, (3) the FYP thesis and defence, (4) the TOEFL score, (5) the university SOPs. A finished thesis with a 300+ GRE and two good Fulbright essays beats a perfect portfolio with a missed April deadline.

14. Scholarship holders, EducationUSA, and how to get your documents read

- **EducationUSA (US Department of State):** the official source on US higher education, with 430+ advising centres in 175+ countries and a "Your 5 Steps to U.S. Study" framework. Find the advising centre in Pakistan through the centre finder (link in Chapter 15). An adviser can read your SOP and résumé, explain the I-20 and visa steps, and tell you what each university actually asks for — use them, they are free.
- **Fulbright alumni / USEFP:** USEFP's own FAQ and advising contacts are the authority for Fulbright questions. Ask USEFP directly (info@usefp.org) about anything the FAQ does not answer — including the parallel-applications question in Chapter 3.
- **Scholarship holders:** ask a Fulbright alumnus or a current US MS student to read one essay and one SOP and mark three things to cut and three to keep. Give them

a week. Do not ask them to write for you, and do not pay anyone for “guaranteed selection”.

- **Message template:** “Assalam-o-Alaikum, I am Muhammad Ismail, a final-year BS Energy Engineering student at UET Peshawar. I am applying for Fulbright (USEFP) and for MS programmes in power systems/energy for Fall 2028, with a final-year project on low-head hydropower from KP irrigation canals. Would you have time to read my Study Objectives essay draft (attached) and tell me the three weakest lines? I would be very grateful. Thank you.”

15. Resources — verified YouTube tutorials and other guides

Every link below was checked on 5 November 2026. Watch them in the order given: start with the two MS-application videos, then the SOP video, then funding, then TOEFL, then the visa walkthrough. Do not binge-watch; watch one, then do the task it describes.

15.1 YouTube tutorials — watch these first

#	Video (click the title)	By	Watch it for
1	Steps on how to apply for MS in USA	iSchoolConnect	The full MS application procedure step by step — where to start, entrance exams vs university choice, what documents go with the application. Watch this first.
2	Fall 2025 MS Application Timeline — Your Ultimate Guide — MS in USA	Gradvine (peer-mentorship platform; mentors are past admits)	A dated month-by-month timeline for a Fall intake, what to do at each stage, common mistakes. Framed for Fall 2025, so its dates are past — the timeline structure and steps still apply to your Fall 2028 plan.
3	Statement of Purpose	UC Berkeley Events (Graduate Diversity programme)	How a US university itself explains the SOP. The video page carries almost no description text, so pair it with the written SOP material in Chapter 8.
4	How to Study in USA for FREE? | Scholarships & Financial Aid Explained	upGrad Study Abroad	Financial aid, need-based and merit scholarships, full-ride options, funding options specifically for master’s students, and research/ teaching assistantships; Fulbright is covered as one route.
5	How I got 119/120 on	Andrew Ahn (scored 119/120)	Section-by-section TOEFL strategy (Reading, Listening, Speaking, Writing) with timestamps,

	the TOEFL test		free resources, and a free study guide linked in the description.
6	My F1 Visa Experience / USA / Approved in a minute / Question asked / Islamabad Embassy	A Pakistani student (PhD in chemistry, on scholarship)	The full Islamabad Embassy walkthrough — compulsory + supporting document checklist, shuttle/entry procedure and on-the-spot fees, the exact questions asked (university choice, application process, professor contact, future research, marital status), a 60-second interview, and the home-ties/return-to-Pakistan tip. Watch this the week before your own interview.

> **Two gaps, stated honestly.** No Fulbright Degree (Pakistan) step-by-step YouTube video could be fully verified — the closest candidates could not be opened (YouTube rate limits) or were about the wrong programme (Fulbright TEA teacher training). No GRE preparation video could be verified either, for the same reason. They are not listed above. Use USEFP's own pages and the ETS official preparation material instead — and treat any "Fulbright video" a friend sends you with the same check: is it the Degree programme, for Pakistan, and from USEFP or an alumnus?

15.2 Other resources — read, don't just bookmark

Resource	Link	Use it for
EducationUSA (official, US Dept. of State)	educationusa.state.gov	The official source on US higher education; "Your 5 Steps to U.S. Study". Advising-centre finder: educationusa.state.gov/educationusa-advising-centers
USEFP — Fulbright Degree FAQ (official)	usefp.org — Fulbright Degree FAQ	USEFP's own answers on eligibility, GRE, essays, references, selection, and the return rule. The application portal itself is usefp.org.
ETS — GRE General Test registration (official)	ets.org — GRE registration	How to register for the GRE (account, test-centre/at-home booking), plus the free official test prep available at registration.
ETS — TOEFL iBT (official)	ets.org — TOEFL iBT about · toefliBT.ets.org — register and practise	The two official TOEFL sites — test information, and the registration/practice portal with free and paid prep.
Written guide — MS in the USA, step by step	iSchoolConnect — 10-step guide	Choosing universities, eligibility (16 years of education), deadlines, exams and scores, essays and résumé, applying. Pairs with video 1 above.

Written guide — Fulbright Pakistan (USEFP) in full detail	StudyTribune — Fulbright guide	Pakistan's USEFP process end to end: GRE at application (minimum 145 per section), ETS code 9388, two essays, three references, April 1 deadline pattern, interviews June–August, 2-year home-residency requirement. Cross-check every date against USEFP's own pages.
Written guide — fully funded master's / RA-TA route	Global Study Board — fully funded master's	The four funding routes (RA, TA, fellowship, external awards), why research-based thesis master's degrees fund best, how to find funded programmes and faculty before applying, and how to read an offer letter's financial terms.

16. Final checklists, tracker, fraud warnings, sources

16.1 Master documents checklist

- **Fulbright (by ~1 April 2027):** online form; three reference letters; GRE score report (ETS codes 9388 + 2326); scanned transcripts including grading-scale explanation (HEC-attested preferable); two essays (Study Objectives + Personal Statement).
- **Direct applications (Dec 2027 – Feb 2028):** passport (valid at least 6 months beyond your intended stay — check expiry now); degree + final transcript (after Oct 2027); GRE score (only where required/optional — check each page); TOEFL/ IELTS score; résumé (Chapter 10); SOP per university (Chapter 8); three LORs (Chapter 9); application fee per university.
- **After admission:** financial documents for the I-20 (Chapter 12.2); SEVIS fee receipt; DS-160; visa interview documents (Chapter 12.3).

16.2 Application tracker — fill in September–October 2027 (2028 dates are not published yet)

Route / Programme	Deadline (with time zone)	GRE?	English rule	Fee	LORs	Submitted?	Result / funding
Fulbright (USEFP)	~1 Apr 2027 (pattern)	Required, min 145/ section	TOEFL only if shortlisted	Free	3 references		
Stanford MS EE							
Berkeley ERG							
MIT TPP							

Michigan MEng ESE							
Wisconsin MS ECE (Research)							
Colorado Boulder MS ECE							
NC State MS-EPSE							
Oregon State MS ECE							
Idaho MS EE							
Montana State MS EE							

16.3 Fraud warnings

- Apply only through USEFP (Fulbright) and each university's official application portal. No agent can apply for you, and no agent can guarantee selection, an RA, or a visa.
- Never pay anyone for "guaranteed Fulbright selection", a "reserved seat", or a "direct admission + funding" package. Never share portal passwords. Never send original documents to a stranger.
- Never copy another scholar's essay or SOP. Universities and USEFP check, and a copied essay ends the application.
- Be careful with "consultants" who charge large fees for university shortlists: the shortlist in Chapter 5 and the EducationUSA advising centres are free.

16.4 Sources and re-check dates

Profile facts: Muhammad Ismail's refined CV (2 pages, November 2026) and his FYP/project records as shared in this conversation; project statuses as of 5 November 2026. Fulbright facts: USEFP Fulbright Degree pages (usefp.org/scholarships/fulbright-degree.cfm and /fulbright-degree-FAQ.cfm) and US State Dept 9 FAM 302.13, opened 5 November 2026. Funding facts: official pages of Michigan ME/ECE/CSE/Robotics, Purdue, Georgia Tech, UT Austin, CU Boulder, SIU, and EducationUSA, opened 5 November 2026. GRE/English facts: ETS official pages (format, scoring, fees effective 1 Aug 2026). University facts: official programme pages listed in Chapter 5, opened 5 November 2026; professor papers verified by DOI. Costs: official bursar/ fee pages and State Department fee schedule, opened 5 November 2026. YouTube and web resources: checked 5 November 2026 (Chapter 15). **Re-check in mid-to-late 2027:** the USEFP 2028 Fulbright call (dates, final-transcript rule, parallel-applications answer); every programme's GRE rule, English minimum (old vs new TOEFL scale), application fee, tuition, and

funding page; the State Department fee schedule (Visa Integrity Fee) and the Pakistan reciprocity table (F-1 issuance fee); ETS fee tables at booking. **No Fall 2028 dates, deadlines, tuition rates or funding figures were published anywhere at the time of writing** — everything dated in this booklet is the 2026–27 cycle, and the tracker is deliberately blank until you fill it from official pages.

Prepared 5 November 2026. Companion to the Erasmus Mundus v2 booklet (34 pages, same date). If a fact in this booklet conflicts with USEFP’s or a university’s official page in the application season, the official page wins — always.